

Team 14 Product Backlog: LitClub

Team Members:

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Problem Statement:

While there are sites online for users to discuss their favorite pieces of literary media, few replicate the experience of going to a book club in person. In a book club, people can discuss novels as they go along; no spoilers are being discussed because the understanding is that everyone has read up to a certain point. Sites like GoodReads only provide spaces to review books completely with or without spoilers; we intend to create a forum-like site for users to discuss books chapter-by-chapter as they go. Here, users can connect with a reading community at their own pace. Additionally, by bringing in these social features, we distinguish our site from StoryGraph's "buddy read" feature by broadening the audience that users interact with. Our app serves as a platform for passionate readers to connect with others who are reading the same books as them, interested in similar genres, and have their own recommendations to share. Along with a chapter-by-chapter reading forum, our app will also have a dedicated social network for users to share updates about what they're reading, share book recommendations, and more. Through our app, readers around the world can connect with one another and discuss their current reads.

Background Information:

Audience: Reading remains one of the most popular forms of entertainment to this day, and many people want to find a community centered around the books that they love. Other apps used to track and review read books only offer a place to review the book as a whole; we aim to provide a platform where readers can record their thoughts as they go along. This will allow users to connect with others without having had to read the entire book first. Our primary audience is people who are currently reading books and want a virtual space to discuss them as they progress.

Similar Platforms: Several similar platforms that we have encountered include Goodreads, StoryGraph, and Pagebound. While these apps have gained popularity in the book lover community, they have varying degrees of functionality. Goodreads, which is by far the most popular, allows readers to submit reviews on books and create a library of books they wish to read. StoryGraph mainly provides data visualization for readers about their habits, including pages, genres, and books read. Pagebound, which is the most similar to our app because of the Reddit-style forums and ability to hide spoiler content from other users. However, our app differs from Pagebound because the forums are specifically divided up by chapter instead of having one forum for the whole book with spoiler tags. Overall, LitClub focuses on dividing books into bite-sized forums by chapter while unifying the community of readers.

Limitations: While other apps allow for certain unique features, such as reviewing a whole book or being able to join a virtual book club, no app compiles these features into one, allowing a user to utilize the features they wish to without using an app different from their peers and fellow readers. While Goodreads, one of the most popular apps, serves as a great tracking app for individuals to compile a list of, and StoryGraph, which requires you to be in a book club and encourages reading the same books and material as your club, are both popular book apps, LitClub offers the individuality to read at your own pace and also allow you to join communities and groups with similar current reads. With LitClub, the reader can truly navigate their reading habits and styles in one app, at their preference and convenience. Additionally, our app allows readers to share and discuss thoughts chapter-by-chapter instead of simply a holistic review at the end.

Functional Requirements:

1. As a reader, I would like to create a user profile with a username, password, profile image, and a nickname.
2. As a reader, I would like to add a short biography to my profile.
3. As a reader, I would like to select what genres of books I'm interested in seeing recommended to me.
4. As a reader, I would like to select books I am currently reading to put on my shelf.
5. As a reader, I would like to return to books on my shelf to go to their forums.
6. As a reader, I would like to bookmark pages/chapters to return to.
7. As a reader, I would like to checkmark chapters to access the forum for that chapter.
8. As a reader, I would like to modify the status of a given book between four statuses: Want to read, Currently reading, On hiatus, or Finished.
9. As a reader, I would like to see books recommended to me based on my previous reading.
10. As a reader, I would like to be able to choose what reading information I share publicly, such as current reads. I should have the option to hide a book from my public bookshelf.
11. As a reader, I would like to be able to follow other readers.
12. As a reader, I would like to view my followers and following users.
13. As a reader, I would like to direct messages to other users.
14. As a reader, I would like to be able to block other users.
15. As a reader, I would like to save or "dogear" books I want to eventually read.
16. As a reader, I would like to create personalized bookshelves depending on genre or a category of my choosing to compile books into specific reading lists.
17. As a reader, I would like to be able to post blog entries onto my main page without those posts being a part of a given forum.
18. As a reader, I would like other readers to be able to view my past book reviews or forum comments from my profile page.

19. As a reader, I would like to view forums that correspond to each chapter of the book I am reading.
20. As a reader, I would like to share my thoughts by making a comment on a forum.
21. As a reader, I would like to make subcomments under any other comment or subcomment.
22. As a reader, I would like to have the ability to upvote/like comments I agree with.
23. As a reader, I would like to be notified when I receive subcomments from other users.
 - Requires remote development build for testing (limited to 30/month)
24. As a reader, I would like to be able to report comments or posts that I find offensive.
25. As a reader, I would like for comments flagged as having spoilers for chapters beyond the scope of the specific forum to be censored or blurred.
26. As a reader, I would like to mark if my chapter review is from a first-time read or a reread.
27. As a reader, I would like to review the book as a whole.
28. As a reader, I would like to mark my review as having or not having spoilers before publishing my review.
29. As a reader, I would like to rate my books out of 5 stars based on several fixed categories to calculate my overall averaged review.
30. As a reader, I would like to be able to have my ratings averaged for my overall book rating.
31. As a reader, I would like to flag reviews posted by other users that have spoilers.
32. As a reader, I would like to add tags for my reviews that describe my feelings on the book.
33. As a reader, I would like to sort and read reviews by tags.
34. As a reader, I would like to see the average rating of a book based on others' reviews.
35. As a reader, I would like to sort and read reviews by rating.
36. As a reader, I would like to choose which of my reviews, if any, will be displayed/spotlighted on my user profile.
37. As a reader, I would like to be able to navigate to someone's profile from the review they posted.
38. As a reader, I would like to be able to submit summaries for a chapter, with the potential for that summary to be displayed at the top of the chapter's forum.
39. As a reader, I would like to upvote or downvote potential summaries to decide which one gets the forum spotlight.
40. As a reader, I would like to be able to join a book club so that I can read books along with others.
41. As a book club leader, I would like to be able to create a book club that readers can join.
42. As a book club leader, I would like to create a list of books for my book club to read.
43. As a book club leader, I would like to select a book for my club to be currently reading.

44. As a book club leader, I would like to break the books up into sections and to create forum threads on my book club site for discussions of certain chunks of the book.
45. As a book club leader, I would like members to be able to suggest and vote on the next book for the club to read.
46. As a book club leader, I would like to make my book club publicly discoverable and joinable by anyone, or choose to make it private and available only via link.
47. As a book club member, I would like to discover publicly available book clubs from the books I'm currently reading.
48. As a book club member, I would like to access discussion pages that my book clubs have set up.
49. As a book club member, I would like to view past books that book clubs have read.
50. As a book club member, I would like to view the future list that a book club plans to read.
51. As a reader, I would like to be able to scroll through available book clubs that are active for a book that I am currently reading.
52. As a reader, I would like to be able to form and name a book club.
53. As a reader, I would like to be able to comment and post within a club discussing a certain book that the group is currently reading.
54. As a reader, I would like to be able to report rude/hurtful comments and users.
55. As a reader, I would like to be able to mark a comment in a book club (including my own), as a spoiler if they have read past the other members of the book club.
56. As a moderator/team member, I would like to add books and their chapters to the catalogue.
57. As a moderator/team member, I would like to add summaries to books for display.
58. As a moderator/team member, I would like to add images of book covers for display.
59. As a moderator of a reading club, I would like to be able to mark a club as a private or public club (private requires moderator to approve join request, public requires no approval).
60. As a moderator/team member, I would like for comments with sensitive content such as hate speech or slurs to be automatically flagged, and kept from being posted.
61. As a moderator/team member, I would like to be notified of reported comments.
62. As a moderator/team member, I would like to be able to delete reported comments.
63. As a moderator/team member, I would like to be able to allow a reported comment to be posted.
64. As a moderator/team member, I would like to be able to restrict users from commenting.
65. As a moderator/team member, I would like to be able to ban a user from the site.
66. As a moderator/team member, I would like to be able to mark a book club as inactive if no one has been active in the club for more than 6 months. (This can be reversed automatically when someone posts/comments in the club group)

Extra Features (If time allows):

67. As a reader, I would like to create summaries for each chapter of a given book and the most popular one (determined by likes) becomes the heading
68. As a reader, I would like to mark where I am in a book via page numbers. (Question: Page number ____ of ____ pages, at ____%)
69. As a reader, I would like to be able to move my reviews/ratings from a different site onto this app with ease when transitioning to this app.
70. As a reader, I would like to access forums generalized to specific series or authors

Non-Functional Requirements:

Performance

- The system shall respond to user requests within 500 ms for 95% of requests under normal load.
- The system shall respond to cached or repeated requests within 200 ms for 90% of cases.
- The system shall support up to 10,000 concurrent users without noticeable performance degradation.
- The frontend shall be designed to facilitate speedy and efficient content retrieval, minimizing client-side rendering times.
- API calls shall be optimized to reduce server load and data usage, including efficient pagination for large data sets.

Scalability

- The backend architecture shall be horizontally scalable to support usage ranging from a small number of users up to 10,000, with server upkeep costs scaling correspondingly.
- The database shall store only data functionally necessary for app functionality, in order to reduce server costs.
- The system shall support adding additional containers or services without requiring downtime.

Development / Architecture

- The frontend shall be developed with React Native to provide iOS and Android support.
- The backend shall include one or more ASP.NET Core Web APIs hosted by Azure Container Apps.
- The backend shall support containerized APIs, automatic scaling, and microservices.
- The system shall use Azure Cosmos DB as the primary document-based database.

- Additional Azure services shall be used as necessary to support core functionality (such as Azure App Insights for monitoring).
- The architecture shall be modular to allow independent updates of the frontend, backend, and database without breaking system functionality.

Security

- The system shall use Azure for backend hosting and storage of user profile and security information.
- The system shall support multi-factor authentication for user accounts. (if time allows)
- The development process shall follow DevSecOps practices, integrating security into each stage of development.
- Sensitive and necessary user data shall be encrypted at rest and in transit.
- Users shall only be able to view information on other users' profiles that has been explicitly marked as public. Any private information shall not be accessible or visible to other users.
- All authentication and authorization processes shall follow industry standards such as OAuth 2.0.

Usability

- The system shall remain launchable and operational at any time between now and the next two years.
- The system shall be distributable via Apple TestFlight, enabling installation through a shareable link.
- The system shall also be distributable via the Google Play Store.
- The user interface shall facilitate quick and intuitive interaction with core services.
- The app shall support accessibility features to accommodate users with disabilities, including screen reader compatibility and color contrast compliance.
- The app shall provide clear error messages and feedback to guide users when issues occur.

Reliability / Availability

- The system shall maintain 99% uptime under normal operating conditions.
- The system shall be able to recover from critical failures within 1 hour.
- Backups of the database shall be taken at least once every 24 hours and retained for a minimum of 30 days.

- The system shall continue functioning in a limited/offline mode if backend services are unavailable.

Hosting / Development Process

- Team members shall be able to access and modify a minimum of 90% of the codebase from a singular GitHub repository.
- The frontend and backend shall be manually testable independent from one another.
- Successful deployments shall be verifiable via integration tests that run automatically when merging into the develop branch.
- Team members shall be able to review code produced by other team members and approve or reject pull requests as appropriate.
- The team shall use Git/version control best practices, and the development tools and techniques applied shall support these practices.
- The CI/CD pipeline shall automatically deploy to a staging environment before production release